

AREA III:

AB Structures and Environment Engineering and Bioprocess Engineering and Allied Subjects

TOPIC 3:

Bioprocess and Food Engineering

3.2 Prepare and evaluate plans, designs, technical specifications and cost estimates, and supervise the operation and maintenance of refrigeration systems and facilities for agricultural products

Number of items: 5-6 questions

Coverage: PAES, cooling load computation, principles, design, and management of agricultural processing system

TOPIC 3:

Bioprocess and Food Engineering

3.5 Apply principles of engineering and science to food manufacturing and operations, including the processing, production, handling, storage, conservation, control, packaging and distribution of food products.

Number of items: 5-6 questions

Coverage: PAES, cooling load computation, principles, design, and management of agricultural processing system

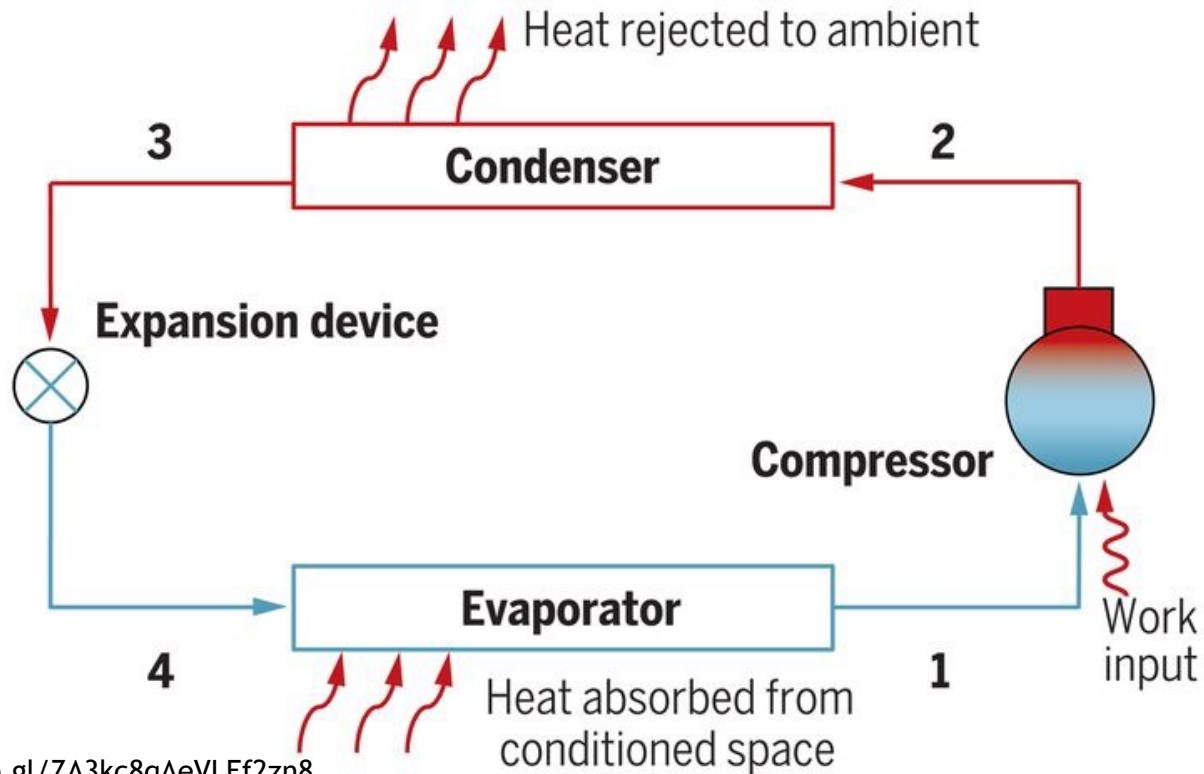
Refrigeration Engineering

This involves the design, analysis, and maintenance of systems that control temperature and humidity, by removing heat from a substance/space.

Encompassing principles of thermodynamics, heat transfer, and fluid mechanics to develop efficient refrigeration system for agricultural products.

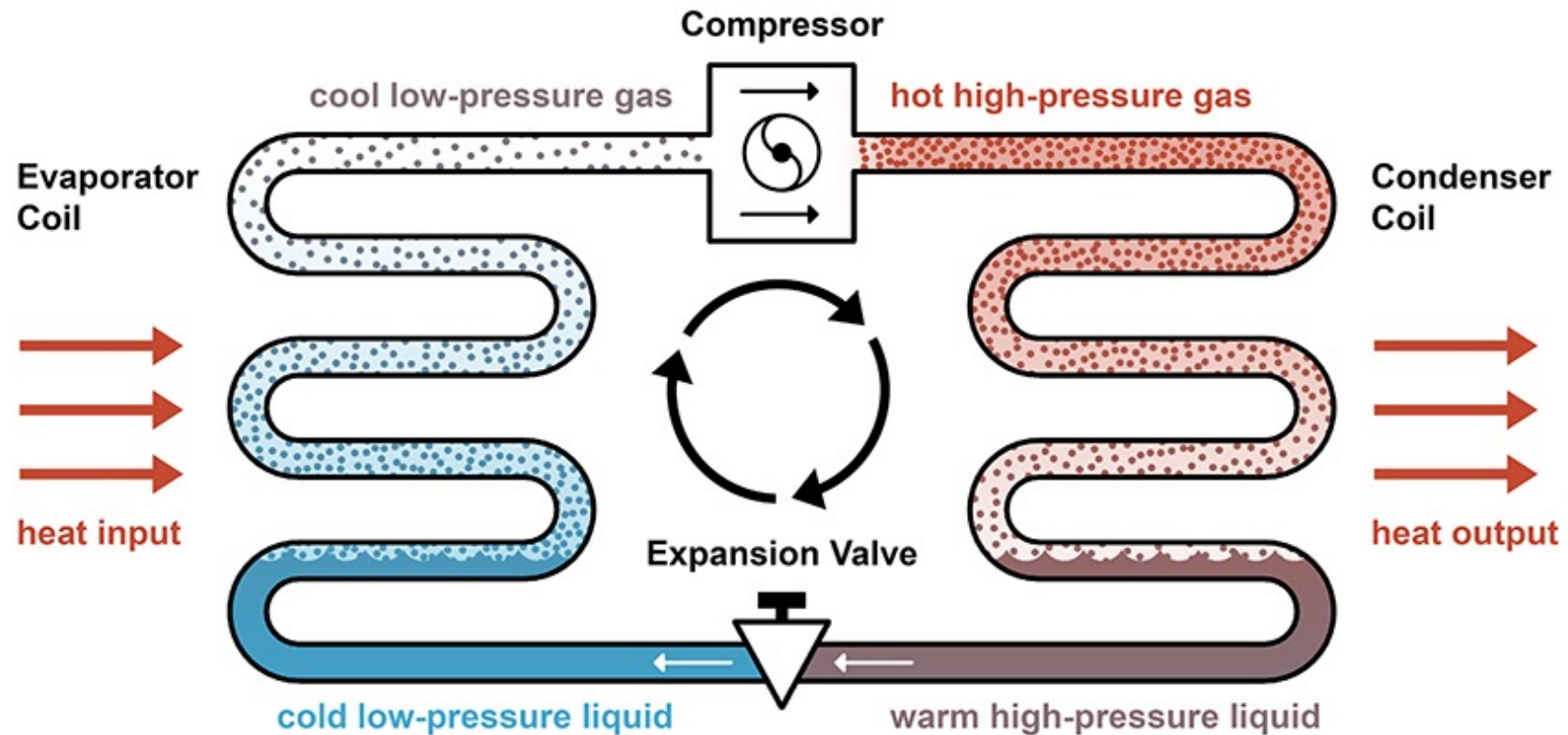
Refrigeration System

Vapor-Compression is known to be the most common refrigeration cycle used in today's time.



Refrigeration System

The Vapor-Compression Refrigeration Cycle



Refrigeration Cycle

Process 4-1: two-phase liquid vapor mixture of refrigerant is evaporated through heat transfer from the refrigerant space.

Process 1-2: vapor refrigerant is compressed to a relatively high temperature and pressure requiring work input.

Process 2-3: vapor refrigerant condenses liquid through heat transfer to the cooler surroundings.

Process 3-4: liquid refrigerant expands to the evaporator pressure.

Refrigerants

Refrigerant Data Including Global Warming Potential (GWP)			
Refrigerant Number	Type	Chemical Formula	Approx. GWP
R-12	CFC	CCl_2F_2	10900
R-11	CFC	CCl_3F	4750
R-114	CFC	$\text{CClF}_2\text{CClF}_2$	10000
R-113	CFC	$\text{CCl}_2\text{FCClF}_2$	6130
R-22	HCFC	CHClF_2	1810
R-134a	HFC	CH_2FCF_3	1430
R-1234yf	HFC	$\text{CF}_3\text{CF}=\text{CH}_2$	4
R-410A	HFC blend	R-32, R-125 (50/50 Weight %)	1725
R-407C	HFC blend	R-32, R-125, R-134a (23/25/52 Weight %)	1526
R-744 (carbon dioxide)	Natural	CO_2	1
R-717 (ammonia)	Natural	NH_3	0
R-290 (propane)	Natural	C_3H_8	10
R-50 (methane)	Natural	CH_4	25
R-600 (butane)	Natural	C_4H_{10}	10

Global Warming Potential (GWP) is a simplified index that estimates the *potential future influence on global warming* associated with different gases when released to the atmosphere.

Refrigerants

► **Chlorofluorocarbons (CFCs) and Hydrochlorofluorocarbons (HCFCs)** are early synthetic refrigerants each containing chlorine. Because of the adverse effect of chlorine on Earth's stratospheric ozone layer, use of these refrigerants is regulated by international agreement.

► **Hydrofluorocarbons (HFCs) and HFC blends** are chlorine-free refrigerants. Blends combine two or more HFCs. While these chlorine-free refrigerants do not contribute to ozone depletion, with the exception of R-1234yf, they have high GWP levels.

► **Natural refrigerants** are nonsynthetic, naturally occurring substances which serve as refrigerants. These include carbon dioxide, ammonia, and hydrocarbons. These refrigerants feature low GWP values; still, concerns have been raised over the toxicity of NH_3 and the safety of the hydrocarbons.

Refrigeration Cycle Keypoints

- Refrigeration capacity is expressed in kW or BTU/hr.
- Alternate unit used is ton of refrigeration which is equal to 200 BTU/min or about 211 kJ/min.
- Coefficient of Performance (COP)
- Carnot Coefficient of Performance
- Heat Pump

Refrigeration Cycle Keypoints

- **Coefficient of Performance (COP)**

COP

A measure of a system's efficiency, calculated as the ratio of heating or cooling output to the energy input. Higher COP means better efficiency.

- **Carnot Coefficient of Performance**

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The theoretical maximum COP based on the Carnot cycle, determined by the temperatures of the hot and cold reservoirs.

- **Heat Pump**

A device that transfers heat from a colder area to a warmer one, used for heating or cooling with high efficiency.

Important Formula...

CARNOT (most efficient)	REVERSE CARNOT (Ref Cycle)	REVERSE CARNOT (Heat Pump)
$Q_A = \frac{W}{\eta}$	$Q_A = (COP)(W)$	$Q_A = (COP)(W)$
$\eta = \frac{T_H - T_L}{T_H}$	$COP = \frac{T_L}{T_H - T_L}$	$COP = \frac{T_L}{T_H - T_L}$
$Q_R = T_L \Delta S$	$Q_R = T_H \Delta S$	* $Q_R = T_L \Delta S$
$Q_A = T_H \Delta S$	$Q_A = T_L \Delta S$	* $Q_A = T_H \Delta S$
$W = \Delta T \Delta S$	$W = \Delta T \Delta S$	$W = \Delta T \Delta S$

Sample questions

Sherwin and Alex decided to live together. As part of their necessities, they decided to buy a new air conditioner. Their chosen air conditioner has an energy efficiency ratio (EER) of 18 kJ/W-h. What is its equivalent COP?

$$\begin{aligned} \text{EER} &= 2.6 \text{ COP} \rightarrow \text{COP} = 6.92 \\ 18 &= 2.6 \text{ COP} \end{aligned}$$

Sample questions

Alex demanded to also buy a refrigerator. As they were looking for one at the mall, they saw a refrigerator that can remove heat from a refrigerated space/substance at -7°C at a rate of 0.37 kJ/s and then rejects it to the environment at 22°C . The minimum required power output is:

- a. 33 W
- b. 40 W
- c. 58 W
- d. 29 W

$$T_H = 22^{\circ}\text{C} \quad Q_R = 0.37 \text{ kW}$$

$$T_L = -7^{\circ}\text{C} \quad W = |Q_R - Q_A|$$

$$\eta = \frac{T_H}{T_H - T_L}$$

$$Q_R = \Delta S T_L \rightarrow 0.37 = \Delta S (-7^{\circ}\text{C} + 273)$$

$$Q_A = \Delta S T_H \quad Q_A = (1.39 \times 10^{-3})(22 + 273)$$

$$Q_A = 0.41 \text{ kW}$$

$$W = |0.37 - 0.41| = 0.04 \text{ kW} = 40 \text{ W}$$

$$\text{COP} = \frac{Q_L}{Q_H - Q_L} = \frac{Q_L}{W}$$

$$\text{COP} = \frac{T_L}{T_H - T_L} = \frac{0.37}{W} = \frac{-7 + 273}{(22 + 273) - (-7 + 273)} = \frac{0.37}{W}$$

Sample questions

Sherwin loves to drink cold water, that is why Alex always puts and refills tap water in a refrigerated water dispenser. The water cools from 29°C to 9°C at a rate of 7 kg/hr. If the COP of this system is 3.33, then the required power input is:

$$T_H = 29^\circ\text{C}$$
$$T_L = 9^\circ\text{C}$$

$$\text{rate} = 7 \frac{\text{kg}}{\text{hr}}$$

$$\text{COP} = 3.33$$

- dit the choices*
- a. 60 W
 - b. 47 W
 - c. 42 W
 - d. 39 W

$$Q = mC_p\Delta T$$

$$Q = \left[7 \frac{\text{kg}}{\text{hr}} \right] \left[\frac{1 \text{ hr}}{3600 \text{ s}} \right] \left[4.186 \frac{\text{kJ}}{\text{kg}\cdot^\circ\text{C}} \right] (29 - 9) = 0.162 \text{ kW}$$

$$\frac{Q_c}{W} = \text{COP} \rightarrow \frac{0.162}{W} = 3.33$$

$$W = 0.049 \text{ kW} = 49 \text{ W}$$

Sample questions



A refrigerator is removing heat from a cold medium at 3°C at a rate of 7200 kJ/h and rejecting waste heat to a medium at 30°C. If the coefficient of performance of the refrigerator is 2, the power consumed by refrigerator is:

- a. 5 kW
- b. 1 kW
- c. 4 kW
- d. 3 kW

$$\text{COP} = \frac{Q_A}{W}$$

$$2 = \frac{7200 \frac{\text{kJ}}{\text{h}} \times \frac{1 \text{ hr}}{3600 \text{ s}}}{W}$$

$$W = 1 \text{ kW}$$

Sample questions

A window air conditioner that consumes 2kW of electricity when running and has a coefficient of performance of 4 is placed in the middle of a room, and is plugged in. The rate of cooling or heating this air conditioner will provide to the air in the room when running is:

edit
the choices

- a. 10 kJ/s
- b. 20 kJ/s
- c. 30 kJ/s
- d. 40 kJ/s

$$\text{COP} = 4$$
$$W = 2 \text{ kW}$$

$$\frac{Q_A}{W} = \text{COP}$$

$$\frac{Q_A}{2} = 4$$

$$Q_A = 8$$

$$W = Q_R - Q_A$$

$$Q_R = 10$$

$$Q_R > Q_A \quad \text{heating}$$

$$Q_A > Q_R \quad \text{cooling}$$

Sample questions

A typical new household refrigerator consumes about 700 kWh of electricity per year and has a coefficient of performance of 1.4. The amount of heat removed by this refrigerator from the refrigerated space per year is:

- a. 3,528 MJ/yr
- b. 2,024 MJ/yr
- c. 9,320 MJ/yr
- d. 1,092 MJ/yr

$$\text{COP} = \frac{Q_L}{W} \rightarrow 1.4 = \frac{Q_L}{\left(700 \frac{\text{kJ} \cdot \text{hr}}{\text{s}}\right) \left(\frac{3600 \text{ s}}{1 \text{ hr}}\right) \text{ year}}$$

$$\begin{aligned} Q_L &= 3,528,000 \text{ kJ/year} \\ &= 3528 \text{ MJ/year} \end{aligned}$$

Sample questions

What is the ideal work for a refrigerator if the cold room temperature is -15°C , ambient temperature is 30°C , and heat rejection is 10kW ?

a. 2.68 kW

b. 5.93 kW

c. 9.30 kW

d. 1.49 kW

$$T_H = 30^{\circ}\text{C}$$

$$T_L = -15^{\circ}\text{C}$$

$$\text{COP} = \frac{(-15 + 273)}{(30 + 273) - (-15 + 273)}$$

$$\text{COP} = 5.73$$

$$5.73 = \frac{Q_A}{10 - Q_A}$$

$$Q_A = 8.51\text{ kW}$$

$$\hookrightarrow Q_R$$

$$\text{COP} = \frac{T_L}{T_H - T_L} = \frac{Q_A}{Q_R - Q_A} = \frac{Q_A}{W}$$

$$Q_R - Q_A$$

$$10 - 8.51$$

$$W = 1.49\text{ kW}$$

$$5.73 = \frac{8.51}{W}$$

$$W = 1.49\text{ kW}$$

Sample questions

The refrigerating capacity of a R-12 system is 22kW. Compressor power is 7.8 hp. Determine the COP of the refrigeration system.

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- a. 2.68
- b. 3.78
- c. 3.94
- d. 2.51

$$Q_A = 22 \text{ kW}$$

$$W = 7.8 \text{ hp}$$

$$\text{COP} = ?$$

$$\text{COP} = \frac{Q_A}{W}$$

$$\text{COP} = \frac{22 \text{ kW}}{7.8 \text{ hp} \times \frac{0.746 \text{ kW}}{1 \text{ hp}}}$$

$$\text{COP} = 3.78$$

Sample questions

One (1) ton of refrigeration is equal to:

- a. 3.52 kW
- b. 1,000 kg
- c. 2,200 lbs
- d. 1 kW

Sample questions

R-22 which is used in residential and commercial air conditioning is also called:

- a. Monochlorodifluoromethane
- b. Dichlorodifluoromethane
- c. Trichlorodifluoromethane
- d. Tetrafluoromethane

Sample questions

R-134a, the green refrigerant which is used to replace R-12 (a CFC) is also called:

- a. Monochlorodifluoromethane
- b. Dichlorodifluoromethane
- c. Trichlorodifluoromethane
- d. Tetrafluoromethane

R410A — combination of
R-32 and R-125

Sample questions

In a refrigeration system, the equipment that rejects heat is the:

- a. Evaporator
- b. Condenser
- c. Compressor
- d. Valve

Sample questions

Considered as the heart of the refrigeration system:

- a. Evaporator
- b. Condenser
- c. Compressor
- d. Valve

THANK YOU!!!

Proverbs 3:5-6

**“Trust in the Lord with all your heart
and lean not on your own
understanding; in all your ways
submit to Him, and He will make
your paths straight”**